

Entity Pulse

Entity Level Sentiment Classification of Financial Data

Milestone – 2

Pavana Lakshmi Gonugunta

G – 579

23BD1A052H

Problem Statement

- Existing financial sentiment models classify entire sentences, ignoring which entity the sentiment refers to.
- This causes inaccurate results when multiple entities appear with mixed sentiments.
- **Example:** "Apple's profits rose, but Tesla's stock fell." →
overall: neutral
entity-level: *Apple*: Positive, ***Tesla*:** Negative.

Our Solution

- Perform **Named Entity Recognition (NER) + Entity-Level Sentiment Analysis**.
- Each entity gets its own sentiment label and confidence score.
- Enables **granular**, interpretable **financial insights**.

How our System Differs

- Based on the **FinEntity** research dataset.
- Uses FinBERT models fine-tuned for financial text.

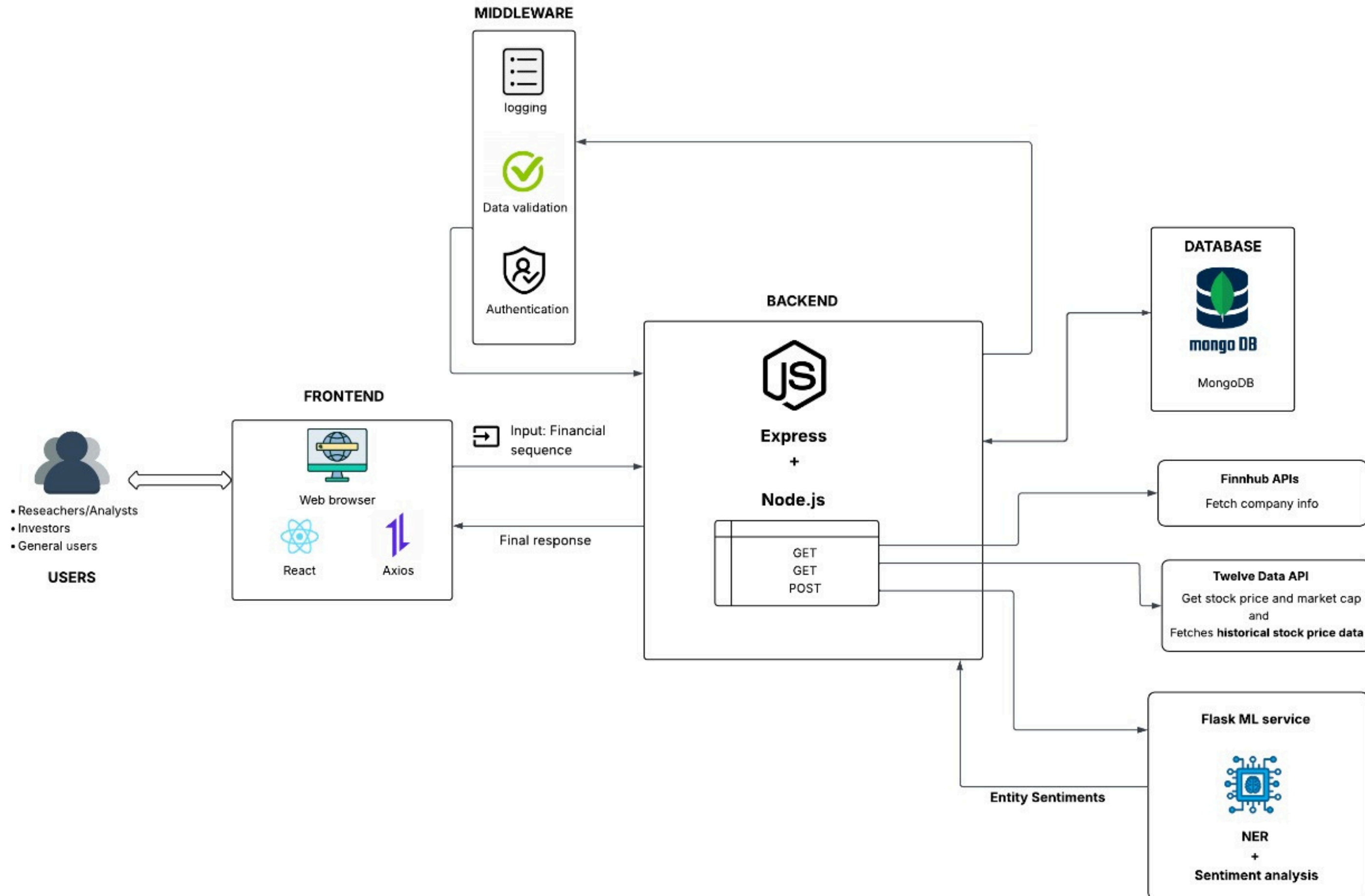
Why it's Better than Existing Datasets:

- **vs. FiQA:** FiQA is aspect-level (e.g., "stock performance" → Negative), but doesn't always link it to the specific entity ("Tesla").
- **vs. SEntiFiN:** SEntiFiN is for short headlines and lacks precise entity tagging.

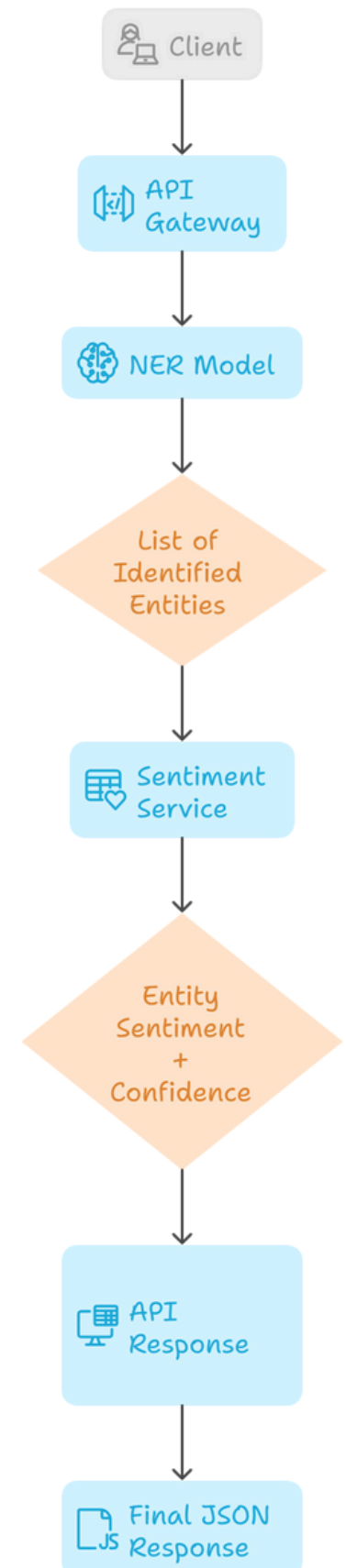
Research Paper Overview

- **FinEntity:** A Dataset for Entity-Level Financial Sentiment Analysis
- Introduced entity-level tagging (**BLOU scheme**) and **entity-specific sentiment classification**.
- Demonstrated that entity-level analysis outperforms traditional sequence-level sentiment tasks.

Architecture Diagram



Sentiment Analysis System Flowchart



Technology Stack

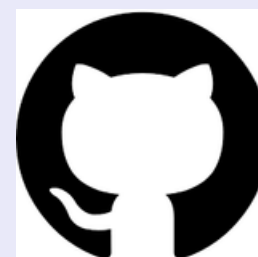
Frontend



Backend



Version Control



Models



Database



External APIs



Model Formulas, Algorithms & Parameters

Key Parameters:

Parameter	Description
max_length	128 for NER, 256 for Sentiment
batch_size	16
learning_rate	2e-5
optimizer	AdamW
epochs	3–5 (Fine-tuning)

Model Significance

- **FinBERT Backbone:** captures domain-specific embeddings.
- **Classification Head:** transforms embeddings into entity/sentiment labels.
- **Softmax Layer:** gives interpretable probability scores.

Softmax Function for Sentiment Probabilities

$$P(y_i|x) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Where z_i = logit for class i (Positive, Neutral, Negative).

→ Highest $P(y_i|x)$ = Predicted Sentiment.

Confidence = max(softmax output).

Cross-Entropy Loss (used during fine-tuning)

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

Ensures model minimizes misclassification.

NER Sequence Labeling Objective

$$\text{Loss}_{NER} = \frac{1}{N} \sum_{t=1}^N CE(y_t, \hat{y}_t)$$

Predicts token-level BIOES labels (B/I/L/U/O).

AdamW (Weight Decay)

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} - \eta \cdot \lambda \theta_t$$

Improved Version of Adam that helps prevent overfitting and improves generalization.

Performance Metrics

NER Model Performance (FinBERT Token Classifier)

- **Validation Loss** reduced from **0.96** → **0.12**
- **Accuracy** reached **96.02%**
- **F1-Score: 0.9596**
- Demonstrates strong entity tagging without a CRF layer.
- Model generalized well — **no overfitting observed.**

NER Model – FinBERT Token Classifier

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1-Score
1	1.7697	0.9634	0.8984	0.8071	0.8984	0.8503
3	0.3785	0.3417	0.9143	0.8929	0.9143	0.8996
5	0.1861	0.2061	0.9311	0.9254	0.9311	0.9240
7	0.1147	0.1415	0.9518	0.9530	0.9518	0.9489
9	0.0730	0.1242	0.9623	0.9634	0.9623	0.9616
10 (Final)	0.0865	0.1210	0.9602	0.9627	0.9602	0.9596

Sentiment Classification Model (FinBERT Sequence Classifier)

Training Summary

- Final Training Loss: ≈ 0.86
- Validation Loss: ≈ 1.24
- Final Accuracy: ≈ 0.82 (81.9 %)
- F1-Score (weighted): ≈ 0.82

Class-wise Evaluation

Sentiment	Precision	Recall	F1-Score	Support
Negative	0.82	0.83	0.83	144
Neutral	0.81	0.79	0.80	160
Positive	0.82	0.83	0.82	162
Weighted Avg	0.82	0.82	0.82	466

Key Learnings, Insights & Applications



Insights from Research Paper

- Entity-level models outperform sequence-level by >10% F1 in multi-entity sentences.
- FinBERT captures financial context better than generic BERT.



Key Learnings

- Token-level annotation quality crucial for accuracy.
- Entity granularity gives actionable insights for investors.
- Importance of integrating real-world APIs for dynamic trend updates.



Real-World Applications

- Investor dashboards showing entity-wise sentiment shifts.
- Early market movement prediction.
- Financial journalism and media analysis.
- Quantitative trading and sentiment-driven portfolio strategies.

Challenges & Limitations

Token-Annotation Alignment Errors

Challenge

- Token-annotation mismatch in BIOES tagging (e.g., sub-tokens like “##ing” misaligned).

Resolution

- Used `return_offsets_mapping = True` and ignored `[CLS]`, `[SEP]`, `[PAD]` tokens to ensure correct entity labeling.

Cross-Language Integration (Node.js ↔ Flask)

Challenge

- CORS and timeout issues between Flask (Python) and Node.js servers.

Resolution

- Enabled CORS in both servers, added JSON error handling, and restricted payload size for stability.

Dataset Format Conversion

Challenge

- FinEntity dataset annotations in span-based JSON not compatible with Hugging Face format.

Resolution

- Built custom preprocessing to map spans into token-level IDs (e.g., B-COMPANY-POSITIVE) with -100 masking for ignored tokens.

Active Development and Planned Improvements

Multilingual Financial Text Support

expanding to handle data in multiple languages.

Aspect-Level Sentiment Analysis

classify sentiment for specific aspects of each financial entity.

CRF Layer Integration

enhance contextual label dependencies for NER.

Ongoing Research Work

We are currently extending Entity Pulse into a **multilingual, aspect-level financial sentiment framework** that can identify **both entities** and their **sentiment** toward **specific aspects** (e.g., product quality, stock performance) across **multiple languages**.



Thank You